INSTRUCTIONS

Follow the directions on the left side of your screen, and respond to the prompts in this sheet. Enter your responses in the white cells and replace "YOUR WORK HERE" with your response to each question.

STEP 0 - Re-Align with Analysis Goal

NOTE: Reference your responses from your Project, Part 1 submission to answer the questions below. You can reuse your accurate responses.

Question 1	**What problem has to be solved? What would be the ideal solution for that problem?** ***TIP:*** **Reference your responses in *Slides A* from your *Project*, *Part 1* submission to answer this question.**
	Listings123 wishes to expand their service to new cities with the proposal to create a marketing campaign to attract new hosts. Johnny currently do not have the budget for the marketing campaign. In order to do so, he needs to convince the investors and VCs for additional funding. To address this problem, it would be important to find data-based evidence indicating what factors make a successful host for Listings123. The ideal solution is to present the data to investors and VCs so that they're confident in offering their funds to go towards the campaign.
Question 2	**What data is important to the problem? What columns are relevant to explore in this EDA?** ***TIP:*** **Reference your responses in *Slides C* from your *Project*, *Part 1* submission to answer this question. To create a new line press *CMD + Enter* (Mac) OR *CTRL + Enter* (Windows).**
	Data Needed: Host Data, Price Data, Listing Data, Location Data, and Reviews Data Host Data- To identify successful hosts. Price Data- To examine if the hosts' prices are competitive. Listing Data- To analyze the demands and availability of the listings. Location Data- To determine the popularity of which the listings are located. Reviews Data- To see if the guests are satisfied.
Question 3	**Is the data enough to solve the problem?**
	The data that was given is not enough information to solve the problem. It would be beneficial if we had more data on the following: What kind of accommodation (vacation rental, house, cabin, etc.), overview of the neighborhood/area, number of bookings/occupants, how long have they been a host since, host response rate, host acceptance rate, the guests reason for reserving, and how many days the guests stayed.

STEP 1 - Explore New and Existing Variables

Question 0	**Which variables in *Wrangled DataExport_2020* are numerical?** ***TIP:*** **Reference your responses in Slides C from your *Project*, *Part 1* or *Part 2* submission to answer part of this question.**

(OPTIONAL)	***Numerical / Quantitative***	Bathrooms, Bedrooms, Beds, Price, Number of Reviews, Days Listed, Review Score Location, Review Score Value
Question 1	**How does *Revenue* relate to the analysis goal/business problem?**	
	Revenue is the total money earned by a listing, for all the days it is listed. This connects to the analysis goal/business problem the investors care primarily about money. This can be the main indicator that the hosts are successful based on how much revenue they have gained and would influence their interest of the investors to pursuing based on that.	
Question 2	**What five <u>numerical</u> variables do you predict will be helpful to explore in relation to *Revenue*? Explain the reasoning as to why you might want to explore each.** ***NOTE:* This is a hypothesis that you will validate throughout this project.**	
	1. Price- Usually in a business, pricing strategies will help create value, profitability, and competition in relations to increase revenue. 2. Number of Reviews- According to most businesses, an increase in reviews relate to an increase in revenue. 3. Days Listed- When the listings been active this means there is more availability to choose, thus it would generate more revenue. 4. Beds- The # of beds can accomodate more consumers which in return would increase revenue. 5. Bedrooms- The # of bedrooms can accomodate more consumers which in return would increase revenue.	

STEP 2 - Group Data Using A Pivot Table		
Question 1	**Based on your *Summary By Host* pivot table, which is the maximum number of listings a single host has? minimimum?**	
	MAX	25
	MIN	1
	Based on your *Summary By Host* pivot table, how many unique hosts are there in your data?	
	19285	
Question 2	**Most of the values for STDEV of Revenue result in error. Why? Is this a useful summary statistic for this data?**	
	There was a division by 0 in the function. No, this may present outliers in the data.	

STEP 3 - Visualize the Distribution of Variables		
Question 1	**Of the numerical variables in *Summary by Host*, which are Left Skewed? Symmetric? Right Skewed? Uniform?**	
	Left Skewed	COUNT of Host ID, AVG of Bathrooms, AVG of Bedrooms, AVG of Beds, AVG of Price, AVG # of Reviews, AVG of Days Listed, AVG of # of Reviews Per Day, AVG of Revenue Per Day, Sum of Revenue, STDEV of Revenue
	Symmetric	AVG # of Amenities
	Right Skewed	AVG of Review Score Value, AVG of Review Score Location
	Uniformly Distributed	N/A
Question 2	**Finalize <u>5</u> variables you will further explore with *SUM revenue* in STEP 4. Explain how your data and visualizations support this selection.** ***NOTE:* They could be similar to the five same variables you identified in STEP 1, Question 2 if your visualizations support those predictions.**	
	1. Beds- It is left skewed similar as the SUM of Revenue histogram. 2. Bedrooms- It is left skewed similar as the SUM of Revenue histogram. 3. # of Reviews- It is left skewed similar as the SUM of Revenue histogram. 4. Price- It is left skewed similar as the the SUM of Revenue histogram. 5. Days Listed- It is left skewed similar as the SUM of Revenue histogram.	

STEP 4 - Examine Relationships Between Variables	
Question 1	**From the 5 variables you had selected, what variables have the strongest relationship with *SUM Revenue.***
	AVG of Number of Reviews because r^2= 0.509. This shows there's a good relationship. AVG of Days Listed has the 2nd strongest r^2 that's equal to .273 when comparing to all the rest of the five scatterplots created.
Question 2	**How do these variable relationships relate back to the analysis goal/business problem?**
	The variables show there are a good relationship in the analysis. This means that this could be evidence of successful host traits to investors along with their interest in money. It relates back to how revenue is important to the business problem as well as the scatterplot showing the relationship with revenue.

STEP 5 - Your Take!	
Question 1	**What additional analysis did you perform? Why?**
	1. I found what city made the most revenue. This would help me find successful host traits by city by city analyze through different variables. To my findings I found San Francisco had the highest sum of revenue. With the highest number of listings (count of host ID) and average of days listed they are attributes that made San Francisco the have the most successful hosts. 2. I analyzed how does room types affect revenue. It would also be helpful in finding another trait and for hosts and investors to notice which room type generates the most revenue. I compared the relationship between room types and average of revenue. During my analysis, I found that hosts that provide the entire home/apt had the highest averge revenue compared to private rooms and shared rooms for the listings. 3. I also uncovered how bed types affect revenue and host success. Again, this could be another finding of a host trait and for the investors to review the revenue of the bed types that contribute to revenue the most. I wanted to see the relationship between bed types to average of revenue. I learned that listings with couches and real beds had the highest revenue opposed to pull-out sofas, futons, and airbeds. Couches had the most highest average revenue. 4. I discovered which amenities led to the most revenue. It would be useful in this analysis to know out of numerous amenities that are being offered by hosts, which ones would be suitable for success. I looked at the relationship between amenities and average of revenue. This resulted to 24 Hour Check-in and private entrance. 24 Hour Check-in had the most highest average revenue.

STEP 6 - Transform Observations to Insights	
Question 1	**Write your insight(s) here. Include *at least* 3 insights.**
	1. Potential hosts in a city with the most number of listings and average of days listed earn the most revenue. (San Francisco has the highest sum of revenue of $10,592,942.31). 2. Potential hosts that offer the entire home/apt earns the most revenue. 3. Potential hosts that offer 24 Hour Check-in and Private Entrance earn the most revenue. 4. Potential hosts that have couches and real beds earn the most revenue. 5. Potential hosts that have high average number of reviews may also earn the most revenue. The hosts can optimize these variables to become succesful and make a lot of revenue. Listings123 should incentify hosts to provide or push for more number of listings, days listed, the entire home/apt, 24 hour check-in, couches, real beds, and number of reviews.